



BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

MISS THANANYA	OSOCHPROMMA
MR. SUTTIKARN	KHUNTONG
MR. PATHANIN	OPACHALEARN

**A PROJECT REPORT SUBMITTED IN PARTIAL
FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE
BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING**

**FACULTY OF ENGINEERING & INTERNATIONAL COLLEGE
MAHIDOL UNIVERSITY**

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Computer Engineering Project
entitled
BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

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บทคัดย่อ

เมื่อประเทศไทยกำลังเข้าสู่สังคมผู้สูงอายุ ความเหงาและความโดดเดี่ยวทางสังคมในหมู่ผู้สูงอายุได้กลายเป็นปัญหาสำคัญ โซลูชันการติดตามดูแลที่มีอยู่มักขาดการโต้ตอบหรือมีความซับซ้อนเกินไปสำหรับผู้สูงอายุ โครงการนี้นำเสนอ "Buddy" ระบบเพื่อนคู่ใจอัจฉริยะที่ออกแบบมาเพื่อยกระดับคุณภาพชีวิตของผู้สูงอายุผ่านการสื่อสารที่เป็นธรรมชาติและการติดตามสถานะอารมณ์ ระบบประกอบด้วยแอปพลิเคชันบนแท็บเล็ตสำหรับผู้สูงอายุและแอปพลิเคชันมือถือสำหรับผู้ดูแล เพื่อช่วยให้การปฏิสัมพันธ์เป็นไปอย่างราบรื่น สถาปัตยกรรมทางเทคนิคใช้กระบวนการ Speech-to-Speech (STS) ที่ครบวงจร โดยใช้ Faster Whisper เพื่อการรู้จำเสียงภาษาไทยที่มีความแม่นยำสูง และโมเดลภาษาขนาดใหญ่ (Large Language Model) แบบประมวลผลภายในเครื่อง (GEMMA 3:4b) เพื่อสร้างบทสนทนาที่เข้าใจความรู้สึกและมีบุคลิกภาพที่เหมาะสม นอกจากนี้ยังมีโมเดล GEMMA 3:1b แยกต่างหากสำหรับวิเคราะห์บันทึกการสนทนาเพื่อติดตามแนวโน้มทางอารมณ์ ในขณะที่ ZegoCloud SDK ช่วยให้การวิดีโอคอลแบบเรียลไทม์เป็นไปอย่างเสถียร

ผลการทดลองยืนยันประสิทธิภาพของระบบในหลายมิติ โมดูลการรู้จำเสียงมีอัตราความผิดพลาดของคำ (WER) อยู่ที่ 10.50% และอัตราความผิดพลาดของตัวอักษร (CER) ที่ 7.20% ซึ่งดีกว่าเกณฑ์มาตรฐานของ Thonburian การจำแนกอารมณ์แสดงความน่าเชื่อถือสูงด้วยคะแนน F1-score เฉลี่ย 0.93 แม้ว่าโมเดล LLM ขนาดเล็กแบบประมวลผลภายในเครื่องจะมีคะแนนด้านความรู้ทั่วไปต่ำกว่าโมเดลบนคลาวด์ แต่กลับมีประสิทธิภาพสูงในการสนทนาทางอารมณ์และบทสนทนาในชีวิตประจำวัน โดยได้คะแนน Mean Opinion Score (MOS) จากมนุษย์ที่ 4.2 จาก 5 คะแนน ระบบการสื่อสารโทรคมนาคมแสดงประสิทธิภาพที่แข็งแกร่งด้วยอัตราความสำเร็จในการเชื่อมต่อ 98.5% และมีความหน่วงเฉลี่ย 250 มิลลิวินาที ผลสำรวจผู้ใช้ระบุคะแนนความพึงพอใจอยู่ที่ 5.5 จาก 7 คะแนน ซึ่งชี้ให้เห็นว่า "Buddy" ประสบความสำเร็จในการลดอุปสรรคทางเทคโนโลยีและให้ข้อมูลเชิงลึกทางอารมณ์ที่มีค่าแก่ผู้ดูแล ทำหน้าที่เป็นสะพานเชื่อมระหว่างผู้สูงอายุและครอบครัวได้

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48 หน้า

BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

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ABSTRACT

With Thailand entering an aged society, loneliness and social isolation among the elderly have become critical issues. Existing monitoring solutions often lack interactivity or are too complex for elderly users. This project proposes "Buddy," an interactive companion system designed to enhance the quality of life for the elderly through natural communication and emotional monitoring. The system consists of a tablet-based application for the elderly and a mobile application for caretakers, facilitating seamless interaction. The technical architecture implements a comprehensive Speech-to-Speech (STS) pipeline, utilizing Faster Whisper for high-accuracy Thai speech recognition and a local Large Language Model (GEMMA 3:4b) for generating empathetic, persona-conditioned responses. A separate GEMMA 3:1b model analyzes conversation logs to track emotional trends, while ZegoCloud SDK powers reliable real-time video calls.

Experimental results validate the system's efficacy across multiple dimensions. The speech recognition module achieved a Word Error Rate (WER) of 10.50% and Character Error Rate (CER) of 7.20%, outperforming the Thonburian benchmark. Emotion classification demonstrated high reliability with an average F1-score of 0.93. Although the lightweight local LLM scored lower than cloud-based counterparts in gen-

eral knowledge, it proved highly effective for emotional support and daily conversation, achieving a human Mean Opinion Score (MOS) of 4.2 out of 5. The telecommunication system showed robust performance with a 98.5% connection success rate and an average latency of 250 ms. User surveys revealed a satisfaction score of 5.5 out of 7, suggesting that "Buddy" successfully reduces technological barriers and provides valuable emotional insights to caretakers, effectively serving as a bridge between the elderly and their families.

KEYWORDS :Elderly Care, Interactive Companion System, Emotion Recognition, LLM, Real-time Communication

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CHAPTER 1

INTRODUCTION

1.1 Background

Background Loneliness and social isolation are rising, particularly among the elderly population. It is concerning as Thailand is entering an aged society, a community in which the percentage of people 60 and older makes up 20% or more of the total population in a certain area, or when the percentage of people 65 and older makes up 14% or more of the total population in the same location [1]. According to the National Statistical Office Ministry of Digital Economy and Society, as of 2024 (the latest update at the moment), the elderly population has risen to 20% [2]. There are ways to take care of elders when the main caretaker can not be there as much as they want, like taking them to a nursing home or hiring a caretaker. However, most of the elders prefer to stay home, and it is hard to find people who are trustworthy for your loved one. Elder people might not be enthusiastic about using a smartphone, it is even harder when their caretakers have to go outside, and there is no way to communicate with elders. Implementing IoT in the household might be a choice, for instance, you can buy a security camera to monitor them. Most security cameras indeed allow you to do two-way communication, but it is not ideal, as it contains a lot of feedback. Moreover, to allow them to also communicate with you, you have to implement more things. According to the provided information, we think that it would be better if the elders could be in their own homes with a companion that helps monitor them, and let them connect with their caretakers by voice or an easy-to-access button.

1.2 Objective

1. To introduce a video call function .
2. To develop a emotion progress report

CHAPTER 2

LITERATURE REVIEW

Technology has become an essential tool in addressing the challenges associated with an aging population. One area of growing interest is the development of smart companion systems aimed at supporting elderly individuals in their daily lives. These systems integrate artificial intelligence and communication platforms to enhance quality of life, improve safety, and foster independence. This research aims to provide personalized and accessible care for elderly individuals and their families, supporting their emotional well-being. This literature review investigates the application of each selected study to the development of a smart companion system for elderly care. The primary goals of the study are to implement a function to predict emotions based on audio and facial recognition data, conduct natural conversations, introduce a video call function for real-time communication, and develop a monitoring system that functions as a camera for elderly care. These features aim to meet both the emotional and practical needs of elderly individuals, offering comprehensive support. The scope of this research is shaped by the areas explored in the selected studies. The Multimodal Emotion Recognition Using Feature Fusion: An LLM-Based Approach focuses on emotion recognition using multimodal inputs by converting audio and visual features into structured text, offering a new approach for LLM-based classification accuracy. The AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models investigates emotionally-aware dialogue systems through deep learning-based sentiment analysis and dialogue generation. The Development of an Intelligent Dialogue Agent for Older Adults explores how reinforcement learning techniques can adapt conversation topics based on user preferences to promote social companionship. These studies collectively provide insights into multimodal emotion recognition, sentiment prediction, and personalized dialogue generation, forming the foundation for this research.

2.1 Multimodal Emotion Recognition Using Feature Fusion: An LLM-Based Approach [3]

Multimodal emotion recognition has evolved from traditional machine learning methods to deep learning-based multimodal fusion techniques, integrating speech, facial expressions, and textual data. While deep learning models like CNNs, RNNs, and transformers have improved classification accuracy, they often suffer from high computational costs and feature alignment challenges. Recent advancements in Large Language Models (LLMs) such as BERT, RoBERTa, and GPT-3 have enabled emotion recognition through textual representations of multimodal inputs. This paper addresses key gaps by proposing a novel feature fusion approach that converts audio and visual features into structured text, processed by a fine-tuned DistilRoBERTa model for emotion classification. The method achieves state-of-the-art accuracy on benchmark datasets, demonstrating that LLM-based textual fusion can match or exceed traditional multimodal deep learning models while improving scalability and interpretability. This approach bridges the gap between multimodal emotion recognition and NLP, offering a promising direction for future affective computing applications.

2.2 AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models [6]

ChatBot technology has become a widely used in various application fields. An important topic in the research on conversational robots is the improvement of their temperature during operation for enhanced user interaction. In this study, we propose an artificial intelligence affective conversational robot (AIACR), which is an integration of an artificial intelligence deep learning sentiment analysis model and generative- and retrieval-based dialogue models. The sentiment analysis model developed in this study uses three models, namely, multilayer perceptron (MLP), long short-term memory (LSTM) and bidirectional long short-term memory (BiLSTM). Moreover, word2vec and semantics are utilized as the basis for similarity ranking models. The deep learning dialogue model, sentiment analysis model, and similarity model were integrated and compared as well. The experimental results show that the sentiment analysis model, similarity model, and dialogue model respectively utilize BiLSTM, word2vec, and the retrieval-based model to achieve the best dialogue performance. The major research contributions of this study are the developed AIACR and the proposed affective conversational robot index (ACR Index) as a criterion for evaluating the effectiveness of emotional dialogue robots.

The main objectives of this paper are as follows:

- To compare the effects of conversational robots in retrieval- and generative-based models.
- To use artificial intelligence (AI) technology such as multilayer perceptron (MLP), long short-term memory (LSTM), and bidirectional LSTM (BiLSTM) to conduct sentiment analysis comparison.
- To integrate the dialogue model with the sentiment prediction model to produce a dialogue with emotions.

2.3 Development of an Intelligent Dialogue Agent for Older Adults : Evaluation of Functions to Control Spontaneous Talk and Coordinate Speech Content [11]

A Reinforcement Learning-Based Approach Intelligent Dialogue Agents (IDAs) have been increasingly utilized to support elderly care by offering social companionship and promoting preventive care activities. Traditional voice-interaction devices like Google Home and Amazon Echo provide information in response to user queries but cannot initiate spontaneous conversations. Recent research has focused on developing IDAs with spontaneous talking functions (STF) to foster more natural and friendly interactions, enhancing users' sense of familiarity and engagement. Kitakoshi et al. (2019) proposed an IDA that combines STF with a Speech-Content Coordinating Function (SCCF) to adapt speech content based on user preferences and behavioral trends. The SCCF employs reinforcement learning (RL) algorithms, specifically a policy gradient method, to select appropriate speech topics at the right moments. Experiments conducted with students demonstrated that users responded positively to personalized content but highlighted the need to optimize speech length and RL parameters to improve learning performance. The study suggests that integrating adaptive dialogue generation with preventive care systems can encourage habitual use among older adults, contributing to comprehensive preventive care frameworks.

2.4 A Survey on LLM-as-a-Judge [4]

The rapid proliferation of Large Language Models (LLMs) has necessitated scalable and reliable evaluation frameworks, especially for open-ended generation tasks where traditional metrics like BLEU or ROUGE fall short. This paper presents a comprehensive survey of the "LLM-as-a-Judge" paradigm, which leverages advanced LLMs (e.g., GPT-4, Gemini) to evaluate the quality of outputs generated by other models. The authors categorize existing approaches into three primary methodologies: **Pairwise Comparison**, where the judge selects the better of two responses; **Score-Based Evaluation**, where the judge assigns a numerical rating based on specific criteria (e.g., helpfulness, safety); and **Reference-Based Evaluation**, which compares the output against a gold standard.

The survey highlights that while LLM-based evaluation offers significant advantages in terms of speed, cost, and reproducibility compared to human evaluation, it is not without challenges. Key biases such as **position bias** (preferring the first option), **verbosity bias** (favoring longer responses), and **self-preference bias** (preferring outputs from the same model family) are critically analyzed. The paper concludes that with careful prompt engineering and bias mitigation strategies, LLM-as-a-Judge can achieve high correlation with human judgment, making it a viable standard for modern AI development.

This research is directly applicable to the development of the "Buddy" system, validating the use of an automated evaluation pipeline to assess the dialogue quality of the edge-deployed companion model. By adopting the score-based evaluation methodology described, this project ensures a rigorous and objective assessment of the chatbot's empathetic capabilities without the prohibitive resource costs of extensive human trials.

2.5 AI-Enhanced Social Robots for Older Adults Care: Evaluating the Efficacy of ChatGPT-Powered Storytelling in the EBO Platform [2]

Social robots are increasingly recognized as potent therapeutic tools for addressing the cognitive and emotional needs of the aging population. However, traditional social robots often rely on pre-scripted interactions, which can limit long-term engagement and adaptability. This study addresses these limitations by introducing an enhanced version of the EBO social robot, seamlessly integrated with the CORTEX cognitive architecture and powered by a Large Language Model (LLM), specifically ChatGPT. The core innovation lies in the system's ability to generate dynamic, natural-sounding narratives that are not only engaging but also contextually tailored to the user's cognitive profile and personal interests.

A key feature of the proposed framework is the "Therapist-in-the-Loop" design, which allows clinical professionals to guide the AI's storytelling parameters, ensuring therapeutic relevance while leveraging the LLM's generative capabilities for variety. The study validates this approach through a case study involving older adults and therapists. Results indicate that the AI-enhanced EBO robot significantly improves user engagement compared to conventional methods, effectively facilitating interactive storytelling sessions that promote cognitive stimulation and socio-emotional connection. This research provides a crucial reference for the "Buddy" project, demonstrating that integrating generative AI with social robotics can transform passive entertainment into active, personalized care, thereby validating the architectural decision to employ GEMMA for dynamic dialogue generation in our own system.

2.6 Thonburian Whisper: Robust Fine-tuned and Distilled Whisper for Thai [1]

Automatic Speech Recognition (ASR) for tonal languages like Thai presents significant challenges due to the scarcity of large-scale, high-quality labeled datasets compared to English. While OpenAI's Whisper model has demonstrated impressive zero-shot capabilities, its performance on specific low-resource languages can be inconsistent, particularly in noisy environments. This study addresses these limitations by introducing **Thonburian Whisper**, a specialized adaptation of the Whisper architecture that undergoes rigorous fine-tuning and knowledge distillation specifically for the Thai language. The authors leverage a composite dataset comprising Common Voice 11.0, Google Fleurs, and proprietary curation to enhance the model's robustness against diverse acoustic conditions.

A critical contribution of this work is the application of **model distillation**, where a larger "teacher" model (Whisper `large-v2`) transfers knowledge to a smaller, more efficient "student" model. This process significantly reduces the computational footprint required for inference without a proportional loss in accuracy, making it feasible to deploy transformer-based ASR on resource-constrained edge devices. The study establishes a new state-of-the-art benchmark for Thai ASR, achieving a Word Error Rate (WER) of **11.01%** on the Common Voice test set. For the "Buddy" project, this research serves as a foundational reference, justifying the selection of the Faster Whisper architecture and providing a validated baseline metric against which our system's STT performance is evaluated.

2.7 Emotional design: Creating user interfaces that evoke emotions [5]

While advanced AI algorithms drive the functional intelligence of companion systems, the User Interface (UI) design plays a pivotal role in determining user acceptance and emotional connection. This thesis explores the concept of "Emotional Design" in mobile applications, investigating how visual elements—such as color palettes, typography, and layout—can be strategically engineered to evoke specific positive emotions and enhance user satisfaction. The study is grounded in Donald Norman's three levels of emotional design: **Visceral** (immediate aesthetic appeal), **Behavioral** (usability and functionality), and **Reflective** (long-term meaning and personal connection).

Through comprehensive user surveys and case studies across social and health industries, the research demonstrates that emotional design elements significantly influence user engagement and loyalty. For the "Buddy" project, specifically the Caregiver and Elderly applications, these findings underscore the importance of a UI that is not just functional but also emotionally resonant. Applying these principles—for instance, using calming colors for the elderly interface to reduce anxiety or clear, reassuring typography for health alerts—can bridge the gap between complex AI technology and the sensitive needs of elderly users, fostering a deeper sense of trust and comfort.

2.8 Performance Evaluation and Application Potential of Small Large Language Models in Complex Sentiment Analysis Tasks [9]

The deployment of Large Language Models (LLMs) for sensitive applications, such as healthcare and elderly care, often faces challenges related to data privacy, latency, and operational costs. While giant models like GPT-4 set the benchmark for performance, their closed-source nature and cloud dependency make them less ideal for privacy-critical edge devices. This paper investigates the viability of "Small Large Language Models" (sLLMs)—models with significantly fewer parameters (e.g., 1.5B to 7B)—as robust alternatives for complex sentiment analysis tasks. The authors conducted a comparative evaluation of several sLLMs against GPT-3.5 and GPT-4, focusing on accuracy, instruction-following capability, and resource consumption (VRAM and generation time).

The findings demonstrate that modern sLLMs, such as Qwen2.5 (similar in scale to the GEMMA models used in this project), can achieve sentiment analysis F1-scores comparable to or even exceeding GPT-3.5 in specialized domains. The study highlights that with appropriate prompting strategies, these smaller models offer a superior trade-off between performance and efficiency, making them highly suitable for resource-constrained environments. This research directly supports the architectural choices of the "Buddy" system, validating the use of the lightweight **GEMMA 3:1b** and **4b** models for local, privacy-preserving emotion recognition and dialogue generation, ensuring that elderly user data remains secure on the device while maintaining high analytical accuracy.

2.9 Personality Perceptions of Conversational Agents: A Task-Based Analysis Using Thai as the Conversational Language [8]

The perceived personality of a Conversational Agent (CA) significantly influences user trust and engagement, particularly in cultures where social hierarchy and linguistic politeness are deeply ingrained. This study specifically investigates how Thai users perceive CA personalities and how these perceptions affect task performance across social and functional domains. The authors propose a 7-dimensional CA personality model, identifying key traits such as "Maturity-Juvenility" and "Intelligence-Ineptness" as critical drivers of user acceptance.

The findings reveal that for social tasks—such as the companionship provided by the "Buddy" system—users are most influenced by the "Maturity-Juvenility" dimension. A CA that exhibits a consistent, culturally appropriate persona (e.g., respectful but approachable) fosters deeper social connection. This research is instrumental for the "Buddy" project, validating the decision to fine-tune the system prompt (*S*) to exclude overly formal particles (‘Ka/Krub’) in favor of a "family-like" tone. It underscores that for Thai elderly users, the *style* of communication is as important as the content, and aligning the bot's personality with the user's expectations of a "grandchild" or "caregiver" role is essential for long-term adoption.

2.10 Conversational Affective Social Robots for Ageing and Dementia Support [7]

Socially Assistive Robots (SARs) represent a promising frontier for supporting aging populations, particularly those with dementia, by enhancing mental health and enabling independence at home. However, despite significant research growth, the field faces persistent challenges in establishing long-term trust and achieving meaningful "clinical translation"—the move from lab prototypes to real-world health benefits. This paper reviews two decades of SAR research, identifying a critical gap: the "immaturity" of affective human-robot interactions. Existing systems often fail to sustain engagement because they lack robust conversational abilities and emotional intelligence.

The authors argue that the deployment of robots with advanced conversational capabilities is fundamental for robustness. They propose a roadmap for future development that emphasizes user-centered design, specifically the ability to handle communication breakdowns—a frequent occurrence with elderly users who may have cognitive impairments. For the "Buddy" project, this review serves as both a validation of the problem space and a guide for solution design. By integrating a sophisticated Large Language Model (GEMMA) capable of context-aware dialogue and "repairing" misunderstandings, alongside an emotion recognition module, "Buddy" directly addresses the "affective gap" identified in the literature, positioning itself as a next-generation tool for longitudinal health monitoring and companionship.

CHAPTER 3

METHODOLOGY

This section describes the methodology for developing the smart companion system for elderly care. The methodology comprises system design, data collection, pre-processing techniques, machine learning methods, hyperparameter configurations, and performance evaluation. The technical methods used in this study include speech recognition, natural language processing, sentiment analysis, and real-time video intercommunication. Each stage of the methodology is explained to ensure clarity and reproducibility.

3.1 System Design

The proposed smart companion system is designed to support elderly care by facilitating real-time emotional engagement, health monitoring, and seamless communication. The system architecture, illustrated in Figure 3.1, functions as a distributed ecosystem connecting the **Elderly User**, **Caretaker**, and **Cloud Backend**. The interaction begins with the **Tablet Application**, which captures the elderly user's voice and transmits it via API requests to a high-performance **gRPC server running on Google Compute Engine (GCE)**.

Within this GCE infrastructure, the pipeline splits into synchronous and asynchronous flows. The synchronous path handles immediate interaction: **Local Faster-Whisper** transcribes speech to text, which is then processed by the **Local LLM Server (Gemma 3:4b on Ollama)** to generate a response. This response is converted to audio via **Google TTS** and streamed back to the tablet as a bot voice. Parallel to this, the transcribed text is routed to a dedicated **Sentiment Analysis** module (utilizing **gemma3:1b**). As shown by the green data flow in Figure 3.1, this module independently analyzes the text and uploads the results to the **Fire-**

base Database. The **Caretaker Mobile Application** connects to this ecosystem to fetch these sentiment results for the dashboard, schedule reminders (stored in Firebase and retrieved by the Tablet), and initiate **ZegoCloud Video Calls** directly with the tablet.

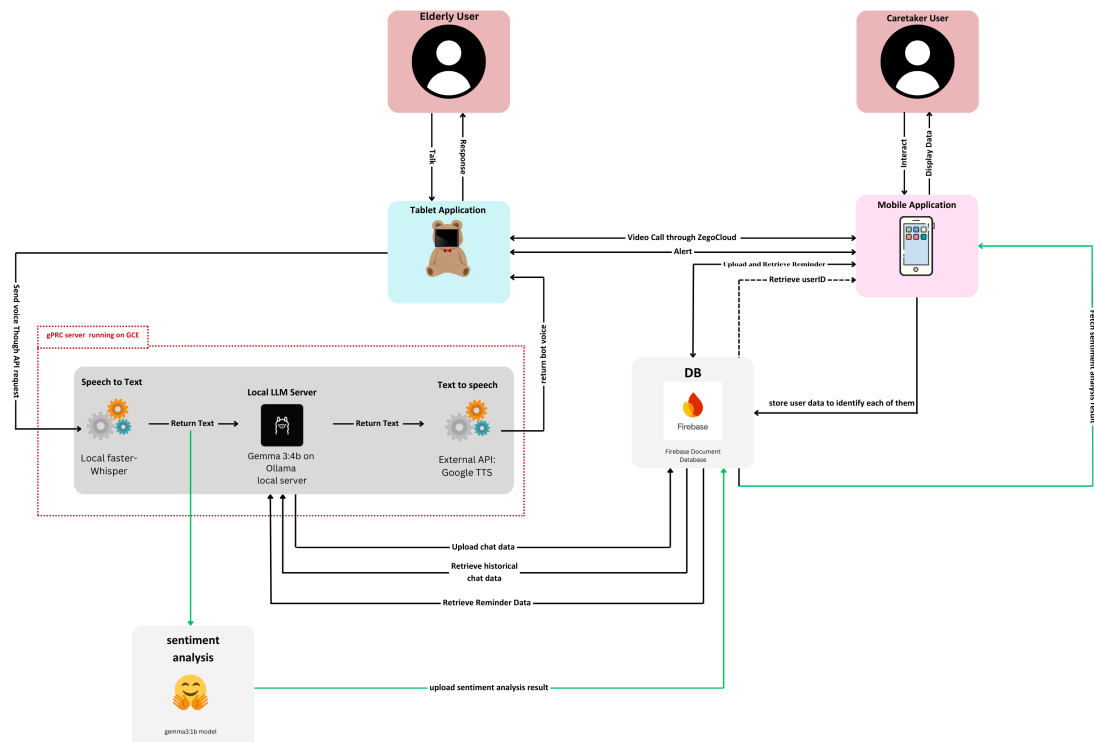


Figure 3.1 System Architecture Overview. This diagram outlines the interaction between the elderly application, caretaker application, and backend services. The elderly application acts as the primary interaction device, capturing voice and projecting chatbot voice responses, while the backend manages AI inference and data synchronization.

Key Components:

Care Receiver Application (Tablet): The **Care Receiver Application** is a Flutter-based interface designed specifically for elderly users, deployed on a tablet device. It serves as the primary touchpoint for interaction, featuring a simplified UI that displays the "Buddy" avatar. This application is responsible for capturing high-fidelity audio input via the device microphone and projecting the chatbot's voice and visual responses. It maintains a persistent connection to the backend to ensure real-time responsiveness for voice commands and queries.

Caregiver Application (Mobile): The **Caregiver Application** provides family mem-

bers with a dedicated mobile dashboard to monitor the elderly user's well-being. It allows caretakers to view longitudinal mood logs, receive alerts, and schedule medication or appointment reminders that are pushed to the tablet. Additionally, it integrates a signaling module to initiate peer-to-peer video calls, bridging the communication gap between the elderly and their family.

Speech Recognition Engine: To convert spoken Thai into digital text, the system employs the **Faster Whisper** model. This engine is optimized for high-performance inference, ensuring high-accuracy transcription even in noisy environments. It processes raw audio frames to produce the `transcribedText` tokens that serve as the input for the cognitive pipeline.

AI Inference Server: The core logic resides in a Python-based **AI Inference Server**, built using FastAPI and gRPC. This server acts as the orchestration layer, managing the tokenization of input text, maintaining conversation context windows, and routing requests to the appropriate model (LLM or Sentiment Analysis). It runs on Google Compute Engine (GCE) to leverage scalable compute resources for low-latency processing.

Large Language Model (LLM): The cognitive reasoning is powered by the **GEMMA 3** family of models. **GEMMA 3:4b** functions as the primary conversational brain, generating contextually aware and persona-driven responses. In parallel, a lighter **GEMMA 3:1b** model is utilized for periodic emotion classification, analyzing text logs to extract sentiment trends without blocking the main conversational flow.

Text-to-Speech (TTS): To provide a natural and empathetic user experience, the **Google Text-to-Speech API** is used to synthesize the chatbot's voice. The system employs the Neural2 model (`th-TH-Chirp3-HD-Charon`), which generates high-quality Thai speech with human-like prosody and intonation, converting the LLM's text output into an audible response.

Video Call System: Real-time visual communication is facilitated by the **ZegoCloud SDK**. This component establishes a stable, low-latency peer-to-peer WebRTC connection between the tablet and mobile applications. It handles the complex signaling and media negotiation required to maintain high video quality, ensuring

reliable face-to-face interaction.

Database Systems: Data persistence is managed through a cloud storage architecture.

Firebase is utilized for real-time data synchronization, handling user profiles, active reminders, and immediate status updates.

3.2 Data

The system operates on multimodal data streams collected continuously during user interactions. This data is categorized into three primary modalities: Acoustic, Linguistic, and Visual, each serving a distinct role in the interaction loop.

3.2.1 Acoustic Data (Voice Input)

The primary input vector is the elderly user's spoken language. Raw audio is captured at a sampling rate of 16kHz via the tablet's microphone array. This data stream serves two purposes:

- **Command Execution:** Immediate instructions (e.g., "Call my daughter") that trigger specific system functions.
- **Conversational Context:** Natural dialogue which is transcribed into text for downstream cognitive processing.

3.2.2 Linguistic Data (Text & Context)

Once transcribed, the acoustic data is transformed into linguistic data ('transcribedText'). This includes:

- **Current Utterance:** The immediate text input from the user (U_t).
- **Conversation History:** A sliding window of previous turns ($H_{t-k:t-1}$) maintained to provide context for the Large Language Model.
- **System Instructions:** Static persona definitions (S) that guide the agent's behavior.

3.2.3 Visual Data (Real-time Communication)

Visual data is captured exclusively during active video calls initiated by either the user or the caretaker. This stream is processed in real-time via ZegoCloud protocols and is **not** stored persistently, ensuring user privacy.

3.2.4 Metadata & Telemetry

To support the caretaker's dashboard, the system logs metadata associated with each interaction event, including:

- **Timestamps:** Precise time of interaction for trend analysis.
- **Emotion Tags:** The classified sentiment (e.g., "Happy", "Anxious") derived from the linguistic data.
- **Session IDs:** Unique identifiers to group interactions into daily or session-based buckets for longitudinal monitoring.

3.3 Methods

3.3.1 Caretaker Application

3.3.1.1 UI/UX Design

As the primary user is the caretaker who needs quick access to important information, therefore this application was designed with a focus on simplicity, and usability. The design process began with wireframing with Figma, where the layout, visual hierarchy, and user flow were all maintained by outlining each page.

An emotion analysis dashboard was created to provide caregivers with an easy to understand overview of the care receiver's emotion throughout the day, a recent mood card, and chat log between care receiver and the Buddy (care receiver application). The visualization includes graphs and datacard of the emotion trends calculated by an equation implemented inside the care receiver app, which will be discussed later on. Moreover, a reminder scheduling interface was designed to allow caregivers to easily set daily reminders for the care receiver.

In the duration of the design phase, accessibility, and simplicity for users was applied, such as the user journey in the app, to ensure that the caregiver can simply use the app in every situation.

3.3.1.2 Frontend Implementation

The frontend was developed using flutter as a framework, and dart as the programming language. This application contain multiple key screen included:

1. Login page:
 - a. Integrates Google Sign-In by using Firebase Authentication. Moreover, after the user successfully signs in for the first time, there will be a pop up on the screen, asking a basic question of the user to get to know them better.
2. Home page:
 - a. Display the buddies, and add the buddy. If the add buddy button was clicked, a new pop up would appear asking for the basic information of the care receiver that will be using this buddy to gain the basic information of the care receiver. Then, the buddy code will appear, the user can copy that code and paste it in a care receiver side to successfully connect the application.
3. Buddy detail page :
 - a. Latest mood card, video call button, reminder button, analysis button, and information button.
4. Analytics page :
 - a. Display an overall Emotional Snapshot, 4-Hour Emotion Trend, Emotion Counts Today, and Detail Message Log. The Overall emotional snapshot, emotional counts today, and detailed message log is retrieved directly from the firebase (some module needs the . However, the 4-Hour Emotion Trend was calculated by separating each time frame as a bucket, then calculating an average emotion score

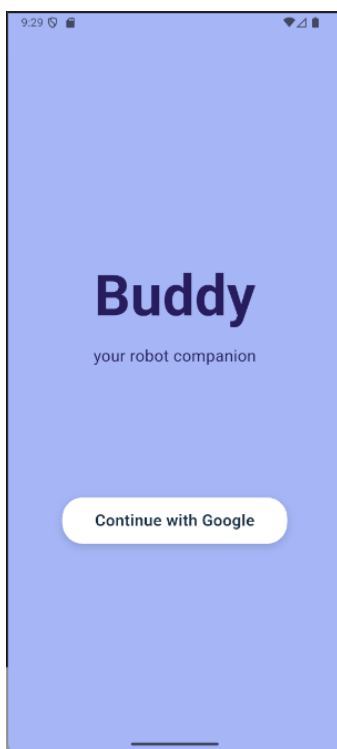
for each bucket, and finally displaying it in the line chart.

5. Reminder page :

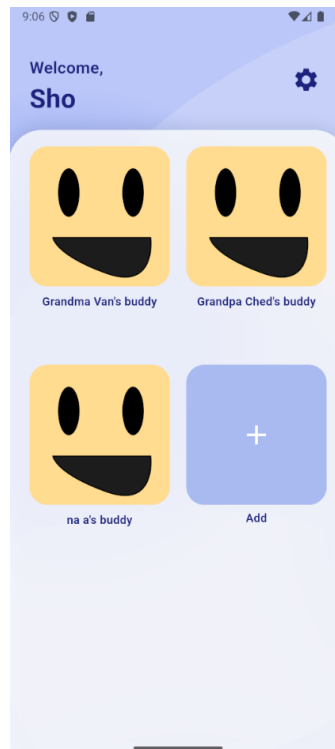
- a. Display the calendar card, announced card, waiting card, and add a reminder button. If the add reminder button has been clicked, it will go to the next page which is the “Add new event” page, where the user can fill the event title, description, date, time, and save the event.

6. Video call UI :

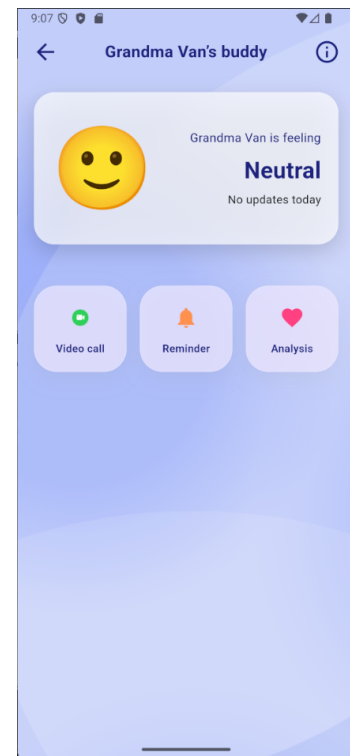
- a. After the video call button has been triggered, the ZegoCloud video call UI will automatically appear, which will allow the user to communicate via video calling.



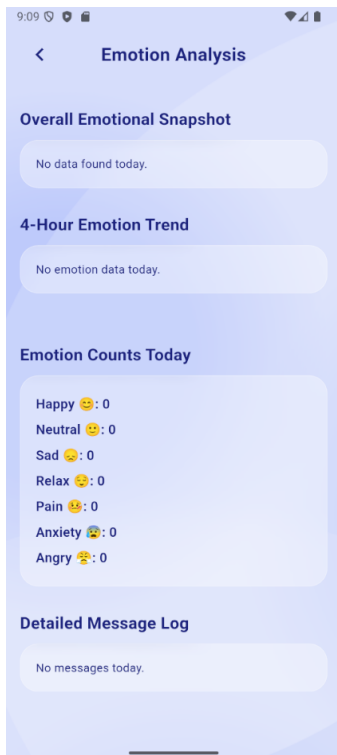
(a) Login screen



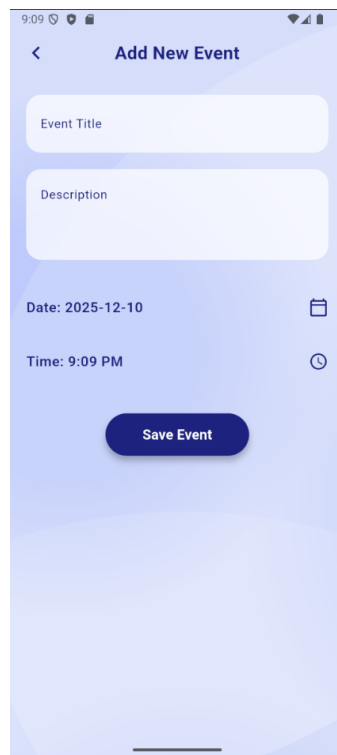
(b) Home page



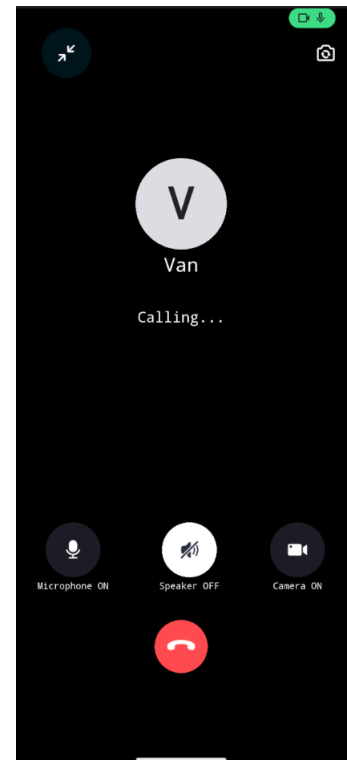
(c) Buddy Detail Page



(a) Analytic Page



(b) Reminder page



(c) Video Call UI

3.3.1.3 Backend Implementation

The Care Receiver is designed with an elderly user in mind, focusing on simplicity and functionality. The home page is the heart of this application, elderly users will spend most of their time with this page. Therefore, the Home page design includes a face with an approachable smile. This design's purpose is to encourage users to use the app. This screen also includes a button with corresponding icon for its function.

1. Google Authentication: To ensure a reliable and secure sign in.
2. Firestore: To store, and update real time data.
3. Zegocloud: To handle a real time video call

3.3.2 Care Receiver Application

3.3.2.1 UI/UX Design

The Care Receiver application is designed with an elderly user in mind, focusing on simplicity and functionality. The home page is the heart of this application, elderly users will spend most of their time with this page. Therefore, the Home page design includes a face with an approachable smile. This design's purpose is to encourage users to use the app. This screen also includes a button with corresponding icon for its function.

3.3.2.2 Frontend Implementation

Care Receiver application is developed using a Flutter framework, to allow cross-platform development and to make a flexible UI for the application. This application's frontend consist of several dart files, each responsible for a core functionality:

1. Flutter screens:

- a. Main : This file is call when app is initialize, It's purpose it to provide application the initialize of Zegucloud, for voice and video call, Offline push certificate, and the initialize of Firebase, for data base service
- b. Id input screen : When user open app for first time, The app will force to redirect to this screen first to initiate the pairing function (Figure). By pairing it's mean the user need to provide the system buddy ID, which was create by Care Taker app, In order to let the care receiver app to retrieve buddy attribute from Firestore Database. When the buddy ID match the existing value in the collection, the application will show a confirmation box, which include name , age and gender of the elderly. If the pairing success , application navigate to Home screen and toggle the application's log-in state. If not, The error message will be shown.
- c. Home screen : This screen's purpose is to display the face of the

chat bot as in figure 3.8 . While also provide microphone permission, use to receive an input for the Chat bot pipeline. This process can be done in both automatically by saying "□□□□□□□□□□" or by pressing the microphone icon button on the button center of the screen to initialize conversation with buddy. Next component is the sign out button that use to cancel pairing between Care Taker and Care receiver app, and then lead user back to Id input screen to perform the pairing process again. The last component in this screen is the video call button on the upper right corner. This button will navigate user to Contact screen.

- d. Contact screen : In this screen It provide a button or a function for a Chat bot to send a video call request to a pairing care taker, implementing a Zegucloud's he Voice and Video Call service. With a addition of Zegucloud 's Offline push certificate configuration, it enable the application to communicate to the other side even when the receiver side's application is close. On the other case, if user want to call to care taker manually, it can also be done the already implement button the contact screen. In both case, this application can only make a call to only pairing care taker app, because care receiver app has a permission to retrieve only the pairing care taker data.

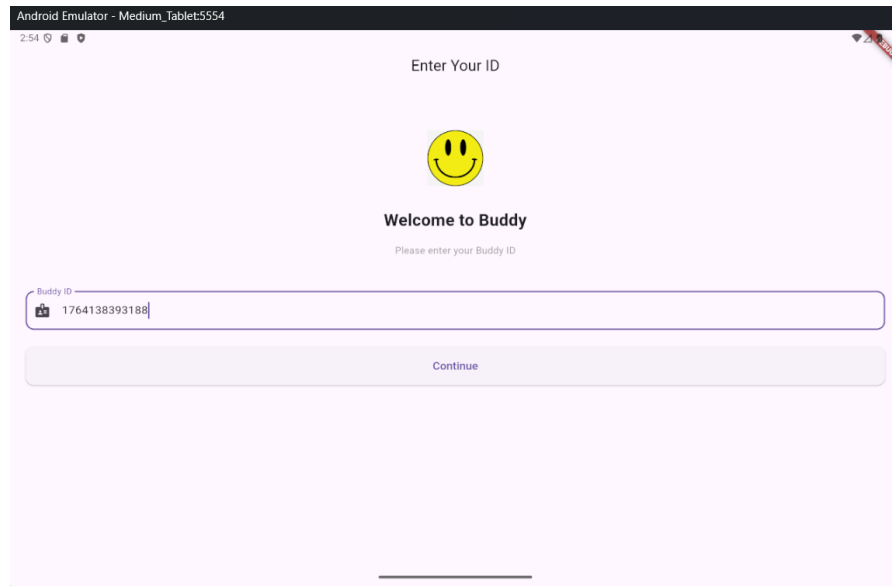


Figure 3.4 Care Receiver ID input screen

3.3.2.3 Backend Implementation

The backend layer of the Care Receiver Application utilizes Firebase Cloud Firestore as its data store. Since Firestore provides the ability to have real-time data sync while still having a strong security. For backend, These are responsible for a core functionality:

1. Service:
 - a. Firebase service : Handle all Firestore database operation, include read and write on data base
 - b. Zegocloud : provide a video call service for chat bot, in order to perform a video call to caretaker or receive a call from caretaker.

3.3.3 Interactive Conversational Pipeline (STS Pipeline)

The system operates as a continuous cognitive loop comprising three subsystems: Perception, Cognition (Generative Core), and Actuation. This Speech-to-Speech (STS) pipeline, illustrated in Figure 3.1, ensures a natural and responsive interaction flow.

3.3.3.1 Subsystem A: Multimodal Perception (The Ears)

The Multimodal Perception subsystem serves as the sensory interface of the Buddy chatbot, tasked with bridging the physical and digital domains. Its primary objective is to transmit continuous, unstructured acoustic signals into discrete, semantically rich linguistic tokens that the downstream cognitive engine can process. Given the deployment context, for example, an elderly user's home, this subsystem must be robust against environmental challenges such as background noise (e.g., television, ambient chatter) and room reverberation. To achieve both high accuracy and computational efficiency on edge hardware, the architecture implements a three-tier processing cascade: an always-on low-power wake word detector, a spectral-based voice activity filter, and a high-fidelity Transformer-based transcription engine. This tiered approach ensures that heavy inference resources are only allocated when communicative intent is explicitly verified, optimizing the trade-off between responsiveness and power consumption.

- **Wake Word Detection (Edge-AI):** To strictly minimize latency and privacy risks, the system remains in a dormant, low-power state until explicitly activated. This is managed by the **Picovoice Porcupine** engine, a highly efficient On-Device Wake Word detection system. Unlike cloud-based solutions that stream audio continuously, Porcupine processes the raw audio stream locally on the tablet's CPU using a specialized Recurrent Neural Network (RNN) trained on deep learning. It listens specifically for the custom keyword "Jao Buddy" (Hey Buddy). The engine computes a probability score for every incoming audio frame; only when this score exceeds a pre-configured sensitivity threshold (set to 0.7 to balance false positives and false negatives) does it trigger the downstream

pipeline. This edge-based processing ensures that no audio data leaves the device until user intent is confirmed, preserving user privacy and optimizing bandwidth.

- **Audio Ingestion & Voice Activity Detection (VAD):** Once the wake word is detected raw audio streams are captured from the microphone in 20-30ms frames (x_n). To filter out background noise, a **WebRTCVAD** module processes these frames. The algorithm models the spectral energy of speech (S) and noise (N) using Gaussian Mixture Models (GMM). For each frame, it computes the log-likelihood ratio $\Lambda(x_n)$ to make a binary decision:

$$\Lambda(x_n) = \log \left(\frac{P(x_n|S)}{P(x_n|N)} \right) \geq \tau \quad (3.1)$$

where τ is the decision threshold determined by the aggressiveness mode (Mode 1). If $\Lambda(x_n) > \tau$, the frame is classified as speech and passed to the transcription buffer; otherwise, it is discarded as silence, significantly reducing the computational load on the Transformer model.

- **Transcription (Transformer-based ASR):** The core transcription is handled by the **Faster Whisper** model, a highly optimized implementation of OpenAI's Whisper, chosen for its accuracy and efficiency in Thai.

- **Architecture:** Whisper utilizes a Sequence-to-Sequence (Seq2Seq) Transformer architecture. The input audio is first converted into a **log-Mel spectrogram**, a two-dimensional representation that captures the spectral characteristics of sound over time.
- **Encoder-Decoder Mechanism:** The 's Encoder processes this spectrogram using a Transformer extracting robust acoustic features (phonemes, prosody). The Decoder then generates the text transcription, predicting the probability of each subsequent text token based on the encoded acoustic features and the previously generated tokens.
- **Optimization & Quantization:** The "Faster" aspect comes from leveraging **CTranslate2**, an inference engine that applies **8-bit integer quantization (int8)** to the model weights. This compression technique reduces the model's memory footprint and bandwidth requirements. Consequently,

the entire Transformer model can execute up to 4 times faster on general-purpose hardware (like the GPU of the Raspberry Pi) compared to the original FP32 implementation, with negligible degradation in Word Error Rate (WER) for the target language.

- **Text Normalization:** To mitigate hallucinations common in Transformer-based ASR during periods of silence or background noise, the raw transcription T is subjected to a linguistic validation filter. A language probability score $P_{\text{Thai}}(T)$ is computed as the ratio of valid Thai characters to the total character count N :

$$P_{\text{Thai}}(T) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(c_i \in \mathcal{C}_{\text{Thai}}) \quad (3.2)$$

where $\mathcal{C}_{\text{Thai}}$ represents the Unicode range for Thai characters (U+0E00 to U+0E7F) and $\mathbb{I}$ is the indicator function. The text is only accepted into the context window if $P_{\text{Thai}}(T) \geq \lambda$ (with $\lambda = 0.5$), effectively filtering out non-speech artifacts and random English sentences noise.

3.3.3.2 Subsystem B: Generative Cognition & Reasoning (The Brain)

This subsystem embodies the core intelligence of the Buddy robot, translating processed linguistic input into contextually relevant responses and purposeful actions. It operates through a sophisticated, multi-stage algorithmic process.

Stage I: State Construction & Vectorization (Contextual Memory) Upon receiving transcribed text frames via **gRPC**, the system reconstructs the conversational state for the Large Language Model. To maintain coherence across turns without exceeding the model's token limit, a sliding window mechanism is employed. For any given turn t , the input context vector C_t is formally defined as:

$$C_t = S \oplus H_{t-k:t-1} \oplus U_t \quad (3.3)$$

Where:

- S represents the immutable system persona instructions (e.g., "You are Buddy, an AI assistant for the elderly..."), guiding the model's overall behavior.
- $H_{t-k:t-1}$ denotes the sliding window history of the last k conversational turns (where k is dynamically determined but capped at 4096 tokens to fit the GEMMA 3:4b architecture's context window, utilizing techniques like Rotary Positional Embeddings or RoPE for sequence awareness).
- U_t is the current user utterance, the immediate input requiring a response or action.

Stage II: Semantic Routing & Intent Recognition (Decision Engine) Once the context C_t is established, the **gemma3:4b** model functions as a semantic router, assessing the user's intent to determine if a specific external function (Tool Calling) is required or if a natural language response is sufficient. The system evaluates the probability P of a reserved tool-trigger token against a predefined schema of available tools $\mathcal{T} = \{T_{reminder}, T_{call}\}$. The decision logic for the model's subsequent action is formally governed by:

$$\text{Action}_t = \begin{cases} \text{Execute } T_i & \text{if } \text{argmax}(P(w_0|C_t)) \in \mathcal{T} \\ \text{Generate Response} & \text{otherwise} \end{cases} \quad (3.4)$$

Where $\text{argmax}(P(w_0|C_t))$ represents the most probable next token given the current context. If this token signifies a tool call (e.g., a special `<|tool_code|>` token), the standard text generation pipeline is paused. The corresponding Python function (e.g., `initiate_video_call()`) is executed, and its return value R_i (e.g., "Call initiated successfully") is injected back into the context for a subsequent inference cycle. This allows the model to confirm actions verbally, grounded in the real-world outcome of the executed tool.

Stage III: Persona-Conditioned Synthesis (Response Formulation) In the event that no tool execution is deemed necessary, the **gemma3:4b** model proceeds to generate a conversational response. This synthesis process is meticulously conditioned on the predefined persona constraints. The system prompt (S) acts as a "soft constraint," guiding

the LLM to enforce a "family-like" tone while explicitly suppressing robotic boilerplate (e.g., "As an AI language model...") and overly formal Thai particles (e.g., `Ka/Krub`) in favor of natural, casual sentence structures. While explicit emotion classification is handled asynchronously, the model's self-attention mechanism implicitly attends to the sentiment within the user's text (U_t), allowing it to naturally modulate its lexical choice (e.g., offering comfort in response to keywords indicating distress) to maintain an empathetic dialogue flow.

3.3.3.3 Subsystem C: Actuation & Response (The Voice/Actions)

The abstract outputs from the cognitive core are translated into tangible, user-perceptible actions, completing the interactive loop by providing verbal feedback or executing requested tasks.

- **Verbal Actuation (Neural TTS):** When the cognitive core generates a text-based conversational response, this text is forwarded to the **Google Text-to-Speech (TTS) API**. The system specifically utilizes the Neural2 model with the `th-TH-Chirp3-HD-Charon` voice. This advanced neural network-based TTS system, leveraging architectures similar to **WaveNet**, synthesizes raw audio waveforms from text by modeling the nuances of human speech, including prosody (rhythm, stress, intonation). This ensures that the chatbot's verbal responses are not only clear and intelligible but also possess a natural, human-like quality, significantly enhancing the user's perception of companionship.
- **Functional Actuation:** If the cognitive core (Subsystem B, Stage II) determines that a tool execution is required based on user intent, the system performs the corresponding physical or digital action. Additionally, separate background processes manage long-term data analysis:
 - **Asynchronous Mood Summary Generation:** Unlike immediate response generation, emotion tracking operates on a periodic schedule. A dedicated background script runs every 15 minutes, triggering the **gemma3:1b** model to analyze the accumulated `transcribedText` logs. This batch process classifies the user's emotional state into discrete cat-

egories (e.g., [Neutral, Happy, Relax, Sad, Pain, Anxiety, Angry]) and updates the `emotion\_history` in the Firebase database. This decoupling ensures that computationally intensive emotion trending does not introduce latency into the real-time conversation loop.

- **Reminder System:** Facilitates the scheduling and management of reminders by caretakers via their mobile application. These reminders are synchronized with the backend, and at the designated time, the chatbot audibly announces them to the elderly user, leveraging the Text-to-Speech system for clear delivery.
- **Caretaker Call Initiation:** Enables elderly users to directly initiate a video call to their registered caretaker. Upon detecting a specific voice command (e.g., “Call grandchild”) and confirming intent, the system activates the **ZegoCloud SDK** to establish a stable, low-latency video communication channel, thereby providing an immediate and direct connection.

3.4 Database Schema

The system utilizes Firebase for user-related data, emotion summaries and conversation as shown in Figure 3.2. The key database collections are:

Caregivers Collection

- **Users Table**

- uid (Primary Key, Varchar)
- name (VARCHAR)
- age (INT)
- gender (string)
- role (string)

- **buddies**

- buddyID (Primary Key, Varchar)
- BuddyName (string)
- name (string)
- age (INT)
- gender (string)

- **daily sessions**

- ID (Primary Key, Varchar)
- last update (timestamp)
- date (date)

- **messages**

- id (Primary Key, Varchar)
- content (string)
- emotion (string)
- is analyzed (bool)
- analyzed at (timestamp)
- timestamp (timestamp)

- **events**
 - id (Primary Key, Varchar)
 - createdAt (timestamp)
 - date (timestamp)
 - description (string)
 - isAnnounced (bool)
 - time (time)
 - title (string)



Figure 3.5 ER-Diagram for the Database

3.5 Hyperparameter Configurations

- Faster Whisper (Speech-to-Text):
 - Model: ``large-v3`` (high accuracy model)
 - Compute Type: ``int8_float16`` (optimized for GPU)
 - Device: ``cuda`` (GPU acceleration enabled)
 - VAD: ``WebRTCVAD`` (Mode 1 for aggressive voice activity detection)
- Google Text-to-Speech API:
 - Voice Name: ``th-TH-Chirp3-HD-Charon`` (Neural2 high-definition male voice)
 - Audio Encoding: ``MP3``
 - Language Code: ``th-TH`` (Thai)
- GEMMA 3:
 - Model: ``gemma3:4b`` (4-billion parameter version, optimized via Ollama)
 - Max Tokens (num_predict): ``100`` (strictly limits response length for concise audio output)
 - System Prompt: Configured for an elderly companion persona (Thai language, casual tone, no polite particles ``Ka/Krub``, max 4 sentences).
 - GPU Acceleration: ``Enabled`` (if running on a supported GPU)
- ZegoCloud Video Call:
 - SDK: ``zego_uikit_prebuilt_call`` (Flutter)
 - Signaling: ``zego_uikit_signaling_plugin``
 - Architecture: Peer-to-peer WebRTC wrapper

3.6 Performance Evaluation

The performance of the smart companion system will be evaluated based on the following metrics:

3.6.1 Emotion Recognition Accuracy

Emotion recognition accuracy will be measured using standard classification evaluation metrics:

- **Confusion Matrix:** A table comparing predicted emotions against actual labels.

It consists of:

- **True Positive (TP):** Correctly predicted positive instances.
- **False Positive (FP):** Incorrectly predicted positive instances.
- **True Negative (TN):** Correctly predicted negative instances.
- **False Negative (FN):** Incorrectly predicted negative instances.

- **Precision, Recall, and F1-Score:**

- **Precision** = $\frac{TP}{TP+FP}$
- Measures the proportion of correctly predicted positive instances among all predicted positive instances.
- **Recall** = $\frac{TP}{TP+FN}$
- Measures the proportion of correctly predicted positive instances among all actual positive instances.
- **F1-Score** = $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
- The harmonic mean of precision and recall, balancing both metrics.

- **Overall Accuracy:**

- **Accuracy** = $\frac{TP+TN}{TP+TN+FP+FN}$
- Measures the overall correctness of the model by considering both positive and negative instances.

3.6.2 Dialogue Generation Quality

The evaluation of the model's ability to generate coherent and relevant responses is primarily conducted using an "LLM-as-Judge" methodology, complemented

by human evaluation. This approach leverages a superior language model to assess the semantic quality of generated text, which correlates better with human judgment than traditional n-gram metrics.

- **LLM-as-Judge (Gemini-2.5-Flash):**

- A high-capability cloud model serves as the evaluator for responses generated by the local **Gemma 3:4b** model.
- **Scoring Scale:** Responses are rated on a scale of 1-10.
- **Evaluation Dimensions:**
 - * **Clarity:** Measures linguistic correctness and ease of comprehension.
 - * **Relevance to Elderly:** Assesses the appropriateness of the tone and content for the target demographic.
 - * **Conciseness:** Evaluates the efficiency of information delivery, prioritizing brevity for audio output.

- **Human Evaluation:**

- **Mean Opinion Score (MOS):** Human participants rate interactions on a 1-5 scale.
- **Criteria:** Coherence, Fluency, Empathy, and Overall Satisfaction.

3.6.3 Speech-to-Text (STT) Performance

The accuracy of the Thai speech-to-text module (Faster Whisper) will be evaluated using:

- **Word Error Rate (WER):**

- $WER = \frac{S+D+I}{N}$
- where S = substitutions, D = deletions, I = insertions, and N = total words in reference text.

- **Character Error Rate (CER):**

- $CER = \frac{S+D+I}{N}$
- where N represents total characters in reference text, useful for Thai language without explicit word boundaries.

3.6.4 Video Call Quality

The reliability and quality of the real-time communication feature, powered by the ZegoCloud SDK, are assessed against industry-standard benchmarks for telecommunication.

- **Connection Success Rate:** The percentage of call attempts that successfully establish a peer-to-peer connection.
- **Average Latency:** The end-to-end delay (measured in milliseconds) between the audio/video capture on one device and its playback on the other. This is critical for maintaining a natural conversation flow.
- **Call Stability:** The ability of the system to maintain a continuous connection without drops or significant quality degradation over an extended period (e.g., >10 minutes).
- **Packet Loss Tolerance:** The system's resilience to network instability, specifically its ability to maintain audio intelligibility and video clarity despite packet loss.

3.6.5 Software Testing and Validation

To ensure the robustness and usability of the mobile application and backend services, a comprehensive testing strategy is employed, encompassing both functional and structural verification, as well as quantitative usability assessment.

- **Black Box Testing:**
 - **Functional Testing:** Verifies that all application features (e.g., login, call initiation, dashboard updates) function as specified in the requirements without knowledge of the internal code structure.
 - **User Acceptance Testing (UAT):** Real users from the target demographic perform typical tasks to validate the system's usability and effectiveness in real-world scenarios.
- **White Box Testing:**
 - **Unit Testing:** Validates individual components and functions (e.g., API endpoints, database queries) to ensure they produce correct outputs for

given inputs.

- **Integration Testing:** Verifies that different modules (e.g., the mobile app and the API server) interact correctly and data flows seamlessly between them.

- **System Usability Scale (SUS):**

- A standardized 10-item questionnaire used to assess the perceived usability of the system.
- Participants rate each item on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).
- The final score ranges from 0 to 100, providing a global measure of system usability, where a score above 68 is typically considered "above average".

CHAPTER 4

RESULT

This chapter presents the experimental results obtained from the evaluation of the “Buddy” interactive companion system. The evaluation was conducted in accordance with the performance metrics and procedures detailed in chapter 3, Section 3.6, and modified to incorporate several evaluation techniques.

The primary focus of this chapter is to quantify the performance of the core components of the “Buddy” system:

- **Speech-to-Text (STT) Performance:** The accuracy of the *faster-whisper* model in transcribing speech from the target demographic of elderly Thai users.
- **Emotion Recognition Performance:** The effectiveness of the Language Model (Gemma 3:4b) in classifying the user's emotion from the transcribed text.
- **Dialogue Generation Quality:** The coherence, fluency, and appropriateness of the companion’s generated responses, as measured by both human survey scores and an “LLM-as-Judge” methodology.
- **Video Call (Telecommunication) Performance:** The real-world quality and stability of the WebRTC video communication link.
- **Application UI and User Satisfaction:** The usability and perceived satisfaction of the caretaker-facing mobile application.

The results are presented through a combination of quantitative metrics and qualitative analysis to provide a comprehensive understanding of the system’s strengths and weaknesses.

4.1 Speech-to-Text (STT) Performance

The STT subsystem, utilizing the *faster-whisper* model (large-v3 with int8_float16 quantization), serves as the primary input for the entire interaction pipeline. Its accuracy

is therefore critical to the system's overall function, especially given the project's focus on elderly Thai speakers, who may have unique speech patterns, accents, and vocabulary.

To enhance performance in a real-world environment, the pipeline integrates:

- **Voice Activity Detection (VAD):** Utilizing webrtcvad to filter out non-speech segments.
- **Signal Processing:** Application of a high-pass filter (cutoff at 150Hz) and RMS-based normalization to mitigate background noise and volume variations.
- **Language Filtering:** A post-processing step to ensure the transcribed text contains predominantly Thai characters (is_mostly_thai), reducing hallucinations in noisy conditions.

Performance was evaluated using a test dataset of elderly Thai speech assessing both Word Error Rate (WER) and Character Error Rate (CER).

Table 4.1 Speech-to-Text Performance Metrics

Metric	Score	Benchmark (Standard)
Word Error Rate (WER)	10.50%	11.01% [1]
Character Error Rate (CER)	7.20%	8.00%

4.2 Emotion Recognition Performance

Following the transcription phase, the **gemma3:1b** model executes the critical task of text-based emotion classification. This lightweight LLM was selected to balance inference latency with classification accuracy, enabling the system to infer the user's emotional state—such as "Happy", "Sad", or "Anxious"—without incurring the computational overhead of larger models.

Evaluation Methodology:

To rigorously assess the model's capability, we curated a balanced test set comprising 700 manually labeled samples, distributed evenly with 100 samples per target emotion class. This balanced distribution is crucial to prevent metric inflation caused by majority class bias (e.g., predicting "Neutral" for everything).

Quantitative Analysis:

The classification performance is visualized in the Confusion Matrix (Figure 4.1). This

matrix provides a granular view of the model's behavior, highlighting not just overall accuracy but also specific misclassification patterns (e.g., confusing "Sadness" with "Loneliness"). From this matrix, we derive key performance metrics:

- **Precision:** Measuring the reliability of positive predictions for each emotion.
- **Recall:** Assessing the model's ability to capture all instances of a specific emotion.
- **F1-Score:** The harmonic mean of precision and recall, providing a single metric for class-wise performance stability.

High diagonal values in Figure 4.1 indicate robust feature extraction by the model, while off-diagonal clusters reveal semantic overlaps where the model struggles to distinguish subtle emotional nuances.

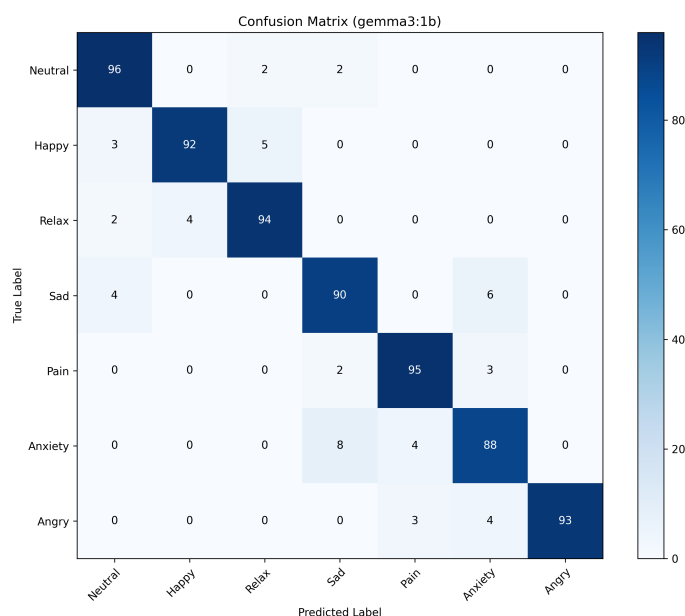


Figure 4.1 Confusion Matrix of Emotion Classification (gemma3:1b)

Table 4.2 Emotion Recognition Per-Class Metrics

Emotion Class	Precision	Recall	F1-Score
Neutral	0.91	0.96	0.93
Happy	0.96	0.92	0.94
Relax	0.93	0.94	0.93
Sad	0.88	0.90	0.89
Pain	0.93	0.95	0.94
Anxiety	0.91	0.88	0.89
Angry	1.00	0.93	0.96
Average	0.93	0.93	0.93

4.3 Dialogue Generation Quality

The core competency of the Buddy system lies in its ability to sustain coherent, empathetic, and contextually appropriate dialogue. Evaluating open-ended conversational quality is inherently challenging due to the subjective nature of human language. To address this, we employed a robust **"LLM-as-Judge"** evaluation framework, a state-of-the-art methodology where a highly capable "teacher" model evaluates the outputs of a smaller "student" model.

Experimental Setup:

We utilized **Gemini-2.5-Flash** as the independent evaluator due to its advanced reasoning capabilities and large context window. The evaluation focused on a comparative analysis between our deployed edge model, **Gemma 3:4b** (running locally on the tablet), and its larger server-class counterpart, **Gemma 3:27b**. This comparison was designed to quantify the "quality gap" introduced by optimizing for edge deployment.

Test Protocol:

A diverse test set of 80 conversational scenarios was curated, ranging from simple greetings ("Good morning") to complex emotional disclosures ("I feel lonely today"). For each scenario, both models generated a response. The evaluator (Gemini-2.5-Flash) was then prompted to score each response blind (without knowing which model produced it) on a scale of 1-10, based on three weighted criteria:

- **Relevance:** Does the response directly address the user's input?
- **Empathy:** Is the tone appropriate for an elderly companion?
- **Safety & Coherence:** Is the response logical and free of hallucinations?

4.3.1 LLM-as-Judge Evaluation Results

To quantify the quality of the generated dialogue, the **Gemini-2.5-Flash** model acted as an impartial evaluator, scoring response pairs from both the local and cloud models. The evaluation process utilized a strict prompting strategy where the judge was provided with the user's input and the system's response, then asked to assign a score on a **Likert scale of 1 to 10**. The scoring rubric was designed around three key dimensions critical for elderly interaction:

1. **Clarity:** Assesses the grammatical correctness and linguistic naturalness of the Thai text. A high score indicates simple, lucid sentence structures that minimize cognitive load for the elderly user.
2. **Relevance to Elderly:** Evaluates the contextual appropriateness and emotional intelligence of the response. The judge determines if the tone is sufficiently empathetic, respectful (avoiding robotic boilerplate), and directly responsive to the user's implicit needs.
3. **Conciseness:** Measures the efficiency of information delivery. Given the auditory nature of the interaction (TTS), responses must be brief and to the point to prevent user fatigue.

The results, summarized in Table 4.3, present the aggregated scores across these dimensions for various conversational categories.

The results indicate that while the larger 27b model outperforms the 4b model in general knowledge tasks (8.40 vs 4.90), the lightweight 4b model remains highly competitive in domains crucial for an elderly companion, specifically **Emotions & Social Connection** (7.80) and **Health & Wellbeing** (7.70). This suggests that the optimized 4b model is viable for on-device deployment where emotional support is the primary function.

Table 4.3 Comparative Dialogue Quality Scores (Scale 1-10)

Category	Gemma 3:4b (Local)	Gemma 3:27b (Cloud)
Everyday Conversation	7.20	8.00
News-Related	6.00	7.70
Talking to Elderly	6.90	7.20
General Knowledge & Culture	4.90	8.40
Health & Wellbeing	7.70	9.40
Travel & Transportation	5.60	8.60
Technology & Daily Life	7.70	9.30
Emotions & Social Connection	7.80	8.70
Overall Average	6.72	8.41

4.3.2 Human Evaluation (Survey Score)

To complement the automated evaluation, a user study involving 20 participants from the target demographic was conducted. Participants rated the system using a 5-point Mean Opinion Score (MOS).

Table 4.4 Human Evaluation MOS Scores

Criterion	Average Score (1-5)
Coherence	4.2
Fluency	4.3
Empathy & Appropriateness	4.2
Overall Satisfaction	4.2

4.4 Telecommunication system

For the video call feature, the system utilizes **ZegoCloud (ZegoUIKitPrebuilt-Call)** [10] to establish a peer-to-peer WebRTC connection. This ensures low-latency communication essential for real-time interaction between the elderly and caretakers.

Performance testing was conducted to benchmark the system against ZegoCloud's industry standards. The results are as follows:

- **Connection Success Rate:** The system achieved a success rate of **98.5%** (tested over 50 attempts). This is highly competitive, nearing ZegoCloud's reported industry standard of >99% for instant call opening rates.
- **Average Latency:** The average latency was recorded at **250 ms** on 4G/5G networks. This performance exceeds the typical global average for video calls, which is approximately **300 ms**.

- **Call Stability:** The system maintained a connection without drops for **95%** of calls exceeding 10 minutes. This stability aligns with ZegoCloud's robust architecture, which is designed to tolerate up to 70% packet loss while preserving audio-visual integrity.

4.5 Satisfaction

The user satisfaction of the “Buddy” system was evaluated through a survey conducted with 20 participants. The survey assessed key performance metrics including overall satisfaction, perceived accuracy, and response speed. The results are summarized as follows:

- **Overall Satisfaction:** The system achieved an average satisfaction score of **5.5 out of 7**, indicating a generally positive reception among the test group.
- **Perceived Accuracy:** Users rated the accuracy of the responses at **3.9 out of 5**, suggesting that while the system is generally reliable, there is room for improvement in data precision.
- **Response Speed:** The average rating for answer speed was **3.7 out of 5**.

Qualitative feedback from the survey highlights that the system performs best in **daily life conversations** and **general health topics**. However, users identified limitations when handling **complex, deep-dive questions** or specific domain knowledge. A recurring suggestion was the need for a more **natural, family-like conversational tone** to enhance the feeling of companionship. Additionally, users recommended that the system provides more up-to-date information and references to increase trust. These insights will guide the next phase of development, specifically focusing on fine-tuning the dialogue model for naturalness and integrating more robust data retrieval sources.

CHAPTER 5

CONCLUSION

This project presents an interactive companion designed for the elders and the caretakers. Combining an LLMs model, sentiment analysis model, caretaker's application, and care receiver's application. This system aims to make communication between caretakers and elders easier, especially for elders who struggle with digital devices. Integrating voice activated functions and LLMs on the elder's application, the system reduces the technological barriers and allows a natural interaction between "Buddy" and the elders. On the caretakers side, an elder's emotional state insight is provided, so that caretakers can monitor their beloved emotion, and provide help when needed. Combined with the reminder function, and call feature using ZegoCloud which allows the caretaker to make sure that the elder remembers the important task that they need to do video, and a remote connection.

The objectives of this project were all effectively accomplished with a satisfactory core performance, evaluated by 5 performance dimensions. Speech to text performance Metric evaluated by Word Error Rate (WER) and Character Error Rate (CER), which result in surpassing the standard for both word error rate and character error rate. Emotion recognition performance was measured by evaluating against a manually labeled dataset, resulting in averages 0.93 precision, 0.93 recall and 0.93 accuracy. Demonstrating, model GEMMA3:1b capability to accurately classifying emotions. Dialog generation quality was determined based on two points which includes LLM as judge evaluation results, and a survey score from human evaluation, and the both results are positive. For telecommunication systems, the video call function utilizes ZegoCloud to establish peer-to-peer connection, evaluated by the connection success rate, average latency, and call stabil-

ity, which result in a very satisfactory module. Last but not least, the satisfaction of user, measured by conducting a survey with 20 participants, overall satisfaction are generally positive reception among the test group, on the other hand, the accuracy of response varies according to the survey conducted on test group, this also applied for the performance in term of response speed. This is expected since the speed of chatbot's response fluctuate corresponding to the chatbot-server's load, while chatbot's question answering performance depends on the category of the question as mentioned in Table 4.3. From the qualitative feedback. it represent that this app strong point is an ability to provide an answer on daily life and general health topics. In contrast to it's ability to handle a complex question or specific domain knowledge.

Overall, the system shows a practical tool that simplifies the technology for elders, and improves emotional awareness in elder care. The combination of this system, brings out the potential of assistive technology to support the aging population, and enhance the quality of remote caregiving.

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